

1a. $m\ddot{x} + b\dot{x} + kx = 0$

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$$

$$F_{\text{damp}} = -b\dot{x} \quad \Delta E_{\text{diss}} = P = F \cdot v$$

$$P = (F_{\text{damp}}) \cdot \dot{x} = -b\dot{x} \cdot \dot{x} = -b\dot{x}^2$$

$$P_{\text{diss}} = b\dot{x}^2$$

+ = gain
- = loss

1b. $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + kx\dot{x} = \dot{x}(m\ddot{x} + kx)$$

$$dE/dt = \dot{x}(-b\dot{x})$$

$$= -b\dot{x}^2$$

$$\hookrightarrow = -b\dot{x}$$

from
 $m\ddot{x} + b\dot{x} + kx = 0$

1c. $P_{\text{diss}} = + \rightarrow \text{Magnitude}$

$$dE/dt = -$$

The time derivative
is the negative value
of the magnitude of P_{diss} .

2a. $y_1(t) = e^{-\beta t} \cos(\omega_1 t)$

$$\lim_{\beta \rightarrow \omega_0} e^{-\beta t} \cos(\omega_1 t) = e^{-\omega_0 t} \cos(\omega_0 t)$$

$$\downarrow \quad \quad \quad \downarrow$$

$$\omega_1 \rightarrow 0 \quad \quad \quad = e^{-\omega_0 t}$$

in the form of

$$A e^{-\beta t}$$

2b. $y_2(t) = e^{-\beta t} \sin(\omega_1 t)$

$$\lim_{\beta \rightarrow \omega_0} e^{-\beta t} \sin(\omega_1 t) = e^{-\omega_0 t} \sin(\omega_0 t)$$

$$\lim = 0 \quad \times$$

$$2c. \lim_{\beta \rightarrow \omega_0} \frac{e^{-\beta t} \sin(\omega_0 t)}{\omega_1} \quad \rightarrow \quad \sin(\theta_{\text{small}}) = \theta$$

$$= \lim_{\beta \rightarrow \omega_0} \frac{e^{-\beta t} (kt, t)}{\omega_1}$$

$$\lim_{\beta \rightarrow \omega_0} e^{-\beta t} \cdot t$$

$$= te^{-\omega_0 t}$$

→
in form
of

$$x_2(t) = Bte^{\beta t}$$

